
Performance Enhancement of Remote Photoplethysmography (rPPG) Systems

A Comprehensive Study on Improving the Accuracy and Reliability of Heart Rate Measurement

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01

introduction

- overview of rPPG
- need for the study

Introduction : Overview of rPPG system

Remote Photoplethysmography (rPPG) is a non-contact technology used to measure physiological signals, such as heart rate and respiration rate, by analyzing subtle changes in

skin color.

1. Basic Principles of Remote Photoplethysmography (rPPG)

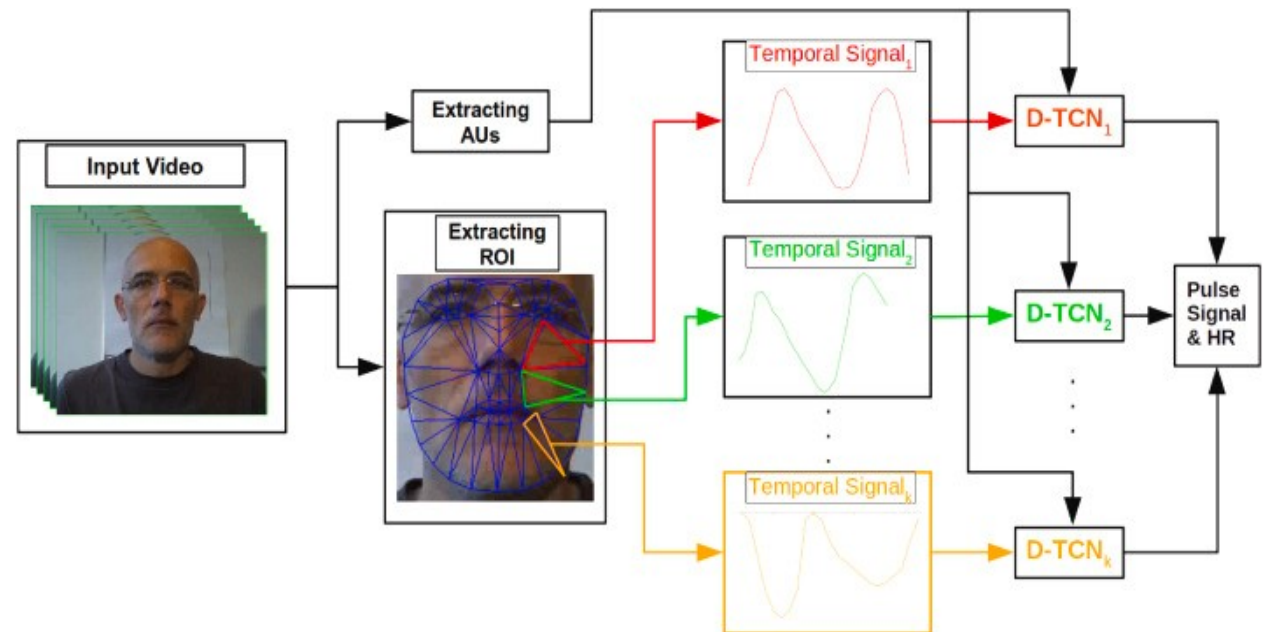
With each heartbeat, blood volume in the skin's microvasculature changes, leading to slight variations in skin color. These color changes are primarily influenced by the oxygen saturation levels of the blood, resulting in detectable fluctuations in specific color channels. Advanced signal processing techniques are applied to accurately derive physiological parameters like heart rate and respiration rate.

2. Relationship Between Skin Color Changes and Physiological Signals

Blood exhibits distinct light absorption properties at different wavelengths. Oxygenated hemoglobin absorbs more red light, whereas deoxygenated hemoglobin absorbs more blue light. These absorption characteristics enable the detection of blood flow changes by analyzing reflected light at specific wavelengths. Consequently, the green color channel is primarily utilized in rPPG systems, as it effectively captures the subtle skin color variations associated with blood volume fluctuations.

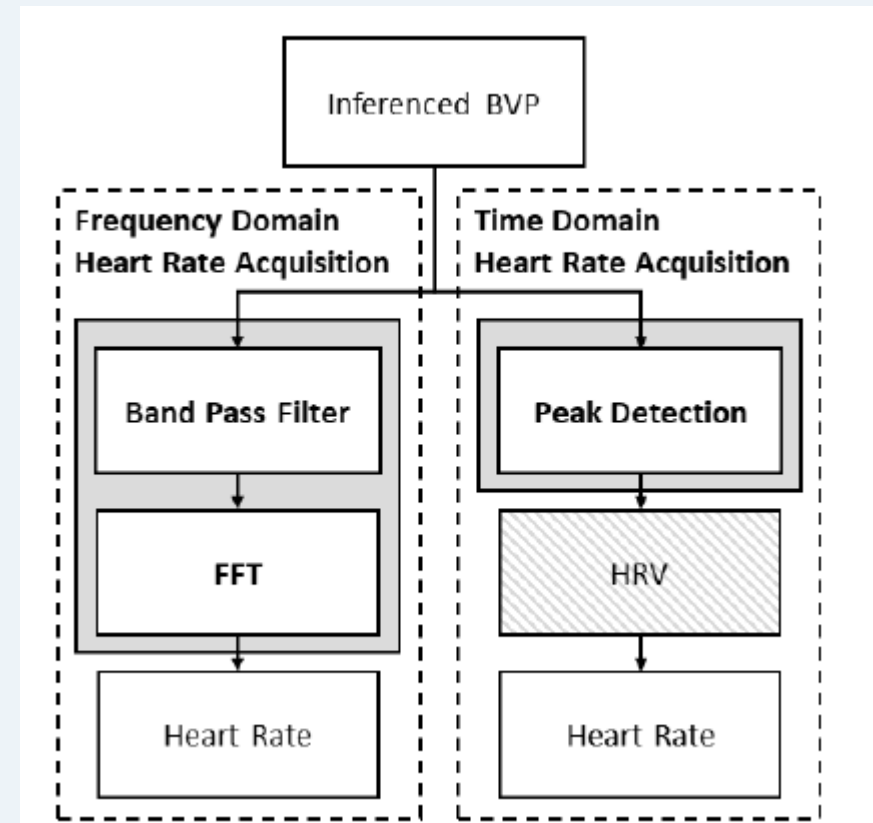
Existing rPPG Technologies

- Non-Deep Learning-Based Techniques
 - Signal Processing Methods
 - Mathematical Models
- Deep Learning-Based Techniques
- Conversion of Extracted Signals to BPM
 - Fast Fourier Transform (FFT)
 - Peak Detection
 - Machine Learning Regression



Current Limitations

- **FFT** : FFT's reliance on specific time windows presents a trade-off: short windows limit frequency resolution, making it difficult to detect subtle changes, whereas long windows improve resolution but increase computational complexity and reduce responsiveness.
- **Peak Detection** : Peak detection in rPPG systems is significantly affected by FPS settings and noise levels.
- **ROI** : single ROI simplifies the system design, it may not consistently provide high-quality signals. On the other hand, fusing data through multiple Regions of Interest (ROIs) is still an area that remains largely unexplored



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Research Hypothesis

- objectives and goals
- methodologies

objectives

1. Develop Advanced Signal Processing Techniques for the error from FFT & peak detection.
2. Develop Multi-ROI Data Fusion Methods.

goals

1. Enhance Accuracy & Reliability.
2. Enable Real-Time Processing.

methodologies

1. Aggregate other frequency domain analysis technics

- Use wavelet transform and welch power spectrum to support FFT results.

2. Integration between FFT and peak detection

To synergize the advantages and mitigate the disadvantages of FFT and peak detection, a holistic integration of their respective data is performed.

3. Selection of ROI

- Regions of Interest (ROIs) with low utility are excluded, and an alpha trimmed mean is applied to the remaining high-reliability ROIs.

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Research Method

- data collection
- measurement
- procedure

Data collection

We utilize the UBFC-rPPG dataset, which records BPM data using PPG sensors.



Table 2: **Benchmark Results.** Performance on the UBFC-rPPG [23], PURE [24] UBFC-Phys [28] and MMPD [26] datasets generated using the rPPG toolbox. For the supervised methods we show cross-dataset training results using the UBFC-rPPG, PURE and SCAMPS datasets.

	Method	Train Set	Test Set							
			PURE [24]		UBFC-rPPG [23]		UBFC-Phys [28]		MMPD [26]	
			MAE [↓]	MAPE [↓]	MAE [↓]	MAPE [↓]	MAE [↓]	MAPE [↓]	MAE [↓]	MAPE [↓]
UNSUPERVISED	GREEN [29]	N/A	10.09	10.28	19.81	18.78	13.55	16.01	21.68	24.39
	ICA [30]	N/A	4.77	4.47	14.70	14.34	10.03	11.85	18.60	20.88
	CHROM [31]	N/A	5.77	11.52	3.98	3.78	4.49	6.00	13.66	15.99
	LGI [34]	N/A	4.61	4.96	15.80	14.70	6.27	7.83	17.08	18.98
	PBV [33]	N/A	3.91	4.82	15.90	15.17	12.34	14.63	17.95	20.18
	POS [32]	N/A	3.67	7.25	4.00	3.86	4.51	6.12	12.36	14.43
SUPERVISED	TS-CAN [7]	UBFC-rPPG	3.69	3.38	N/A	N/A	5.13	6.53	14.00	15.47
		PURE	N/A	N/A	1.29	1.50	5.72	7.34	13.93	15.14
		SCAMPS	4.66	5.83	3.62	3.53	5.55	6.91	19.05	21.77
	PHYSNET [6]	UBFC-rPPG	9.36	17.84	N/A	N/A	5.51	7.47	10.23	12.46
		PURE	N/A	N/A	1.63	1.68	5.07	6.37	13.21	14.73
		SCAMPS	20.08	31.27	4.39	4.39	7.28	9.98	21.05	24.69
	DEEPPHYS [35]	UBFC-rPPG	5.54	5.32	N/A	N/A	6.62	8.21	17.49	19.26
		PURE	N/A	N/A	1.21	1.42	8.42	10.18	16.92	18.54
		SCAMPS	3.95	4.25	3.10	3.08	4.75	5.89	15.22	16.56
	EFF.PHYS-C [36]	UBFC-rPPG	5.47	5.39	N/A	N/A	4.93	6.25	13.78	15.15
		PURE	N/A	N/A	2.07	2.10	5.31	6.61	14.03	15.31
		SCAMPS	10.24	11.70	12.64	11.26	6.97	8.47	20.41	23.52

MAE = Mean Absolute Error in HR estimation (Beats/Min), MAPE = Mean Percentage Error (%).

Measurement

We conduct a comparative analysis of the Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) reported for the UBFC-rPPG dataset in existing studies.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

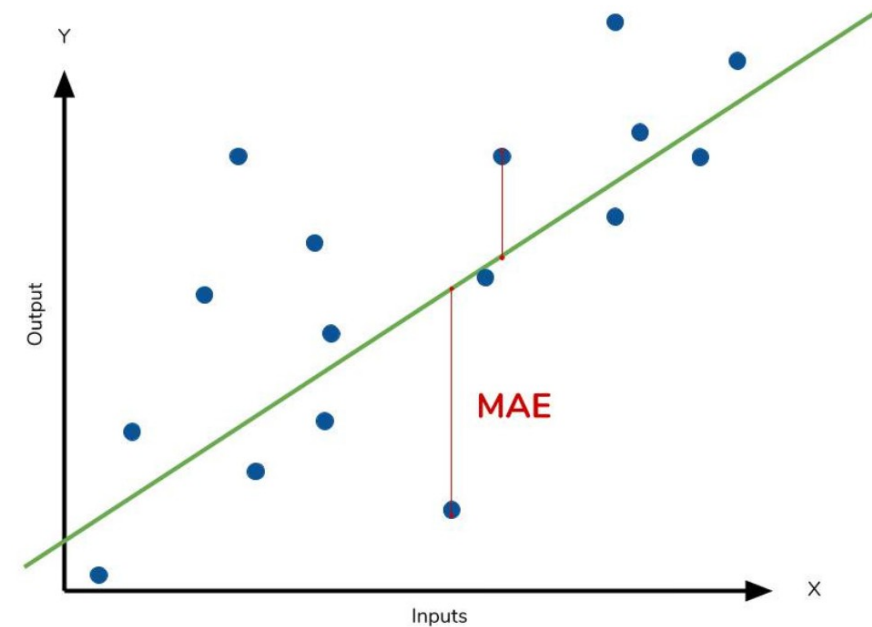
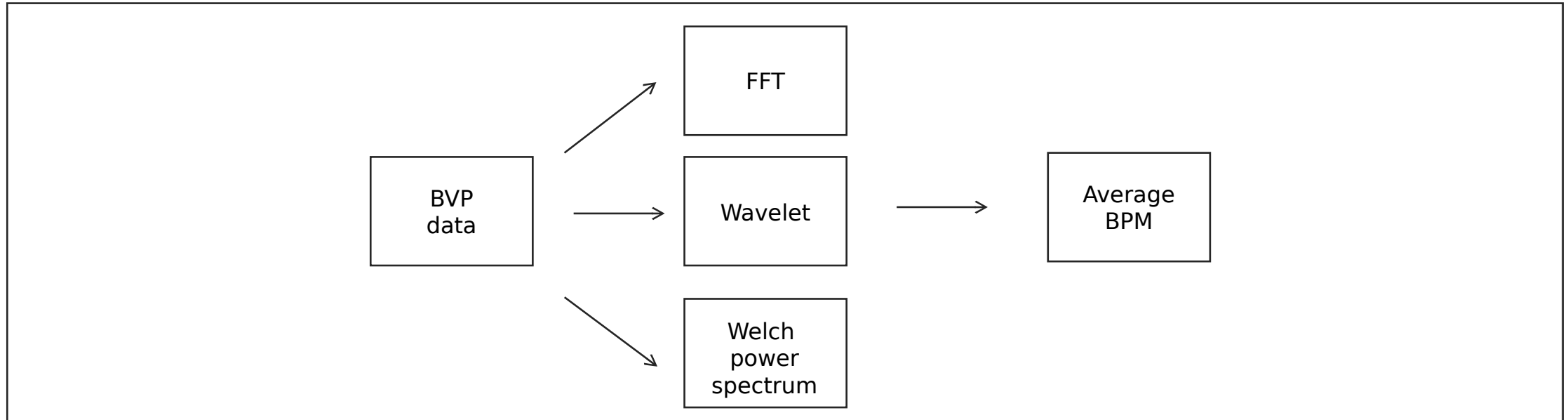


Figure 2 _ Procedure - Frequency domain



- To overcome the limitations of traditional Fast Fourier Transform (FFT), Fast Fourier Transform (FFT), Wavelet Transform, and Welch Power Spectrum are applied to Blood Volume Pulse (BVP) signals, and the average of the three resultant values is calculated

Procedure – Time domain

BPM values derived from peak detection and frequency domain analysis are recalculated by computing a weighted average based on the differences from the previous heart rate to obtain a new BPM value.

Peak Detection HR: 72 BPM

Frequency Domain HR: 71 BPM

Previous HR: 70BPM

Average: New BPM = $(72 \times 0.4) + (71 \times 0.6) = 28.8 + 42.6 = 71.4$

Procedure – Selection of ROI

Among the 21 Regions of Interest (ROIs), three ROIs that exhibit the most similar results in both frequency and time domains are selected based on clustering. An alpha trimmed mean technique is then applied to calculate the average BPM values for each domain

Procedure – Supplementary Signal Processing

- Gaussian Blur
 - Low-pass filter for noise reduction and image quality improvement.
- Savitzky-Golay Filter
 - Data smoothing while preserving signal periodicity.
- Signal Differentiation
 - Precise detection of peaks and rapid changes
- Histogram Equalization
 - Enhances contrast and visibility of subtle signal variations

04

Research Results

- research results
- discussion

Research Results

research

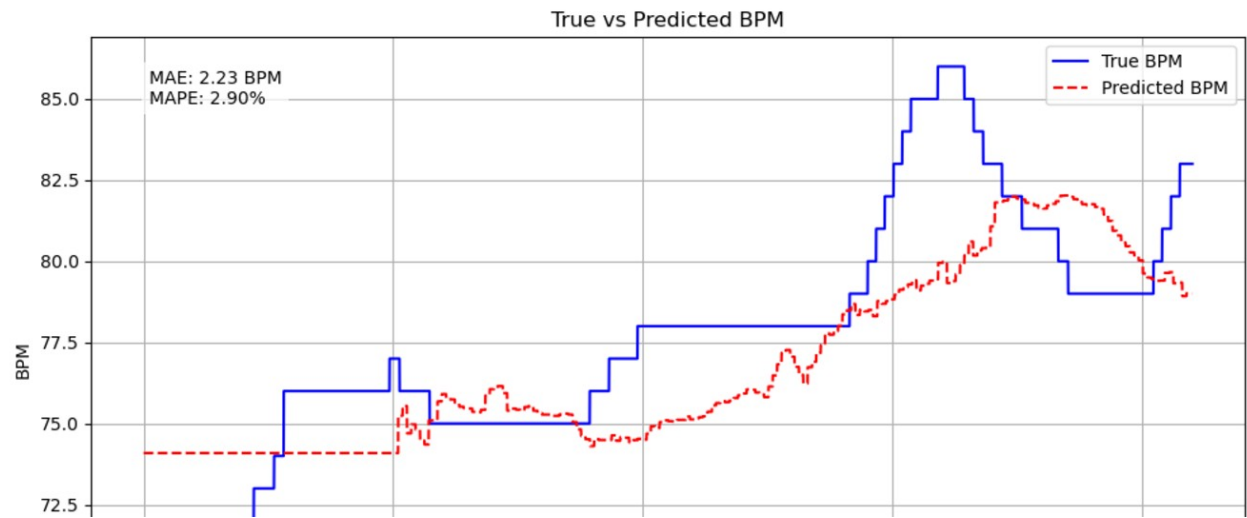
In the case of the Green method, which utilizes the pulse from the G channel as the Blood Volume Pulse (BVP) without applying additional mathematical and signal processing techniques, performance was significantly enhanced. Specifically, the Mean Absolute Error (MAE) was reduced from 19.81 to 2.23, and the Mean Absolute Percentage Error (MAPE) decreased from 18.78% to

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MAE = Mean Absolute Error in HR estimation (Beats/Min), MAPE = Mean Percentage Error (%).

MAE: 2.23 BPM
MAPE: 2.90%



discussion

1. The use of algorithms incorporating mathematical and signal processing techniques resulted in slight performance improvements or, in some cases, a decline in performance.
2. The reliance on multiple ROIs increases computational complexity, potentially hindering real-time processing capabilities. Additionally, while the alpha trimmed mean effectively reduces the impact of outliers, the selection of optimal weights remains subjective and may vary across different datasets and conditions
3. The study predominantly focuses on the UBFC-rPPG dataset, and the generalizability of the findings to other datasets warrants further investigation

Thank you

Thank you for listening to our
presentation